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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fourth Semester B.Tech (CBCS) Degree Examinations

May 2025

AC2004-1 – COMPUTER OPERATING SYSTEM
(For AC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **One full** question from each **Unit**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks****1. What is an Operating System?**

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|---|---|
| A) A collection of programs that manages hardware resources | B) A system software that provides user interface |
| C) A programming language | D) None of the above |

2. What are the main goals of an Operating System?

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|-------------------------------|----------------------------|
| A) Convenience and efficiency | B) Security and networking |
| C) Speed and hardware design | D) None of the above |

3. The Kernel is responsible for

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|-----------------------|--------------------------|
| A) Hardware control | B) User interface design |
| C) Compiling programs | D) None of the above |

4. Batch OS is used in

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|----------------------------|--------------------------|
| A) Single-user systems | B) Multi-tasking systems |
| C) Early computing systems | D) None of the above |

5. A Monolithic OS has

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|------------------------|----------------------|
| A) Single large kernel | B) Layered design |
| C) Microkernels | D) None of the above |

6. The monolithic OS structure means

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|---------------------------------|--|
| A) It is divided into layers | B) It runs all services in a single module |
| C) It does not support hardware | D) It is a virtual OS |

7. Monolithic kernels are mostly used in

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|----------------------|--------------------|
| A) Real-time systems | B) Desktop OS |
| C) Embedded systems | D) Cloud computing |

8. The process of switching from one running process to another is called

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|-----------------------|----------------------|
| A) Interrupt handling | B) Context switching |
| C) Threading | D) None of the above |

9. The process control block (PCB) does not contain

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|------------------|---------------------|
| A) Process ID | B) Program counter |
| C) CPU registers | D) Network settings |

10. The five states of a process include

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|---|---|
| A) New, Ready, Running, Waiting, Terminated | B) Ready, Running, Executing, Terminated, Restarted |
| C) Start, Process, Execute, Stop, Restart | D) Initiate, Execute, Sleep, Wake, End |

11. The key role of the OS in process management is to:
 A) Control and allocate CPU resources B) Manage user input
 C) Design applications D) Manage power settings
12. Inter-process Communication (IPC) mechanisms include
 A) Shared memory B) File transfer
 C) System reboot D) Process termination
13. What is dynamic memory allocation?
 A) Assigning memory at compile-time B) Assigning memory at runtime
 C) Allocating memory using ROM D) None
14. Contiguous memory allocation requires
 A) Continuous blocks of memory B) Random access to memory
 C) Paging D) Virtual memory
15. Paging avoids
 A) External fragmentation B) Internal fragmentation
 C) CPU scheduling D) File corruption
16. What is the main advantage of virtual memory?
 A) Faster execution speed B) More processes can be executed than the physical memory can hold
 C) Reduced CPU usage D) Increases hard disk performance
17. What is demand paging?
 A) Loading all pages of a process into memory before execution B) Loading pages only when they are needed
 C) Loading pages in sequence D) Pre-loading the entire program into memory
18. Which of the following is a valid page replacement policy?
 A) LRU (Least Recently Used) B) FIFO (First-In-First-Out)
 C) Optimal Page Replacement D) All of the above
19. What happens when a page fault occurs?
 A) The operating system loads the required page into memory B) The process terminates immediately
 C) The CPU is reset D) The memory is cleared
20. What is the purpose of a page table?
 A) It maps virtual addresses to physical addresses B) It stores all process control information
 C) It organizes files on disk D) It manages process scheduling

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the concept of turnaround time in a batch processing system.	8	L*2	1	1
	b) Describe the fundamental components of an operating system.	8	L2	1	1
2.	a) Using an appropriate timing diagram, illustrate the priority assignment rule in multiprogramming systems.	8	L2	1	1
	b) What are the common tasks performed by operating system? Explain briefly.	8	L2	1	1
3.	a) Explain partition and pool based resource allocation strategies.	8	L2	2	1
	b) Explain the structure of a kernel-based operating system and describe its key functions and services.	8	L2	2	1
Unit – II					
4.	a) Define child processes and explain their advantages.	8	L2	3	1
	b) Describe the concept of threads in the Solaris operating system.	8	L2	3	1

5. a) Explain the different process states and their transitions. 8 L2 4 1
 b) Discuss the different fields of the Process Control Block (PCB). 8 L2 4 1
6. a) Briefly discuss the relationship between threads and processes. 8 L2 4 1
 b) Define threads. Explain: i) User-level threads ii) Kernel-level threads. 8 L2 4 1

Unit – III

7. a) Explain the concept of paging with a neat diagram. 8 L2 5 1
 b) Consider the page reference string:
 1,0,7,1,0,2,1,2,3,0,3,2,4,0,3,6,2,1 for a memory with three frames.
 Determine the number of page faults using the FIFO, Optimal, and LRU replacement algorithms. Which algorithm is most efficient? 8 L3 5 1
8. a) Compare static and dynamic memory allocation techniques. 8 L2 5 1
 b) Implement Round Robin scheduling with a time quantum of **3 ms** for the following processes:

Processes	P1	P2	P3	P4	P5
Arrival Time	0	1	2	3	4
Service Time	5	3	1	2	3

Find (i) Gantt Chart (ii) waiting time (iii) Average waiting time (iv) Average Turn around time.

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
